Grade 12 Unit 1 Physics Application of physics in other fields

1.1 Physics and other sciences

At the end of this section, you will be able to:

- **explain the relationship of physics with chemistry, biology, geology and astronomy**

1.2 Physics and engineering

At the end of this section, you will be able to:

- **Relate the Newtonian mechanics with civil engineering**
- **List different concepts of physics used in mechanical engineering**
- **Relate electromagnetism to electrical and electronics engineering**
- **Explain how technology contributes to the development of physics**

1.3 Medical physics

At the end of this section, you will be able to:

- **Explain the relation of physics with medicine**
- **Explain how the magnetic property of water is used in MRI for diagnosis**
- **Differentiate the working principle of conventional x-ray and computer
tomography (CT) scan**
- **Discuss how sound wave is used for diagnosis**
- **Explain how radiation is used for cancer and tumor treatment**

1.4 Physics and defense technology

At the end of this section, you will be able to:

- **List different defense technologies**
- **Explain how physics is used in radar, missile and infra-red detection for night vision**

1.5 Physics in communication

At the end of this section, you will be able to:

- **Explain the working principle of different communication technologies.**
- **Explain the relation of physics and communication technology**

Grade 12 Physics Unit 2 Two-dimensional motion

2.1 Projectile motion

At the end of this section, you will be able to:

- **Explain the motion of the projectile with respect to the horizontal and vertical components of its motion.**
- **Derive equations related to projectile motion.**
- **Apply equations to solve problems related to projectile motion**

2.2 Rotational Motion

At the end of this section, you will be able to:

- **Describe the motion of a rigid object around a fixed axis.**
- **Derive equations of motion with constant angular acceleration.**
- **Apply equations to solve problems related to rotational motion.**

2.3 Rotational Dynamics

At the end of this section, you will be able to:

- **Define the physical concept of torque in terms of force and distance from axis of rotation.**
- **Define the physical concept of moment of inertia in terms of point mass and distance from the axis of rotation.**
- **Express torque in terms of moment of inertia and angular acceleration.**
- **Solve problems involving torque and rotational kinematics.**

2.4 Planetary motion and Kepler’s laws

At the end of this section, you will be able to:

- **Describe the motion of the planets around the Sun.**
- **State Kepler’s three laws.**
- **Apply equations to solve problems related to orbital motion.**

2.5 Newton’s law of universal Gravitation

At the end of this section, you will be able to:

- **Explain what determines the strength of gravity.**
- **Describe how Newton’s law of universal gravitation extends our understanding of Kepler’s laws.**
- **Apply equations to solve problems related to Newton’s law of universal gravitation.**